(ii) $A = \bigoplus_{i=1}^n (\dim M_i) P_i$.

(iii) Any indecomposable finitely generated projective module over A is isomorphic to P_i for some i.

Proof. Recall that $A/\operatorname{Rad}(A) = \bigoplus_{i=1}^n \operatorname{End}(M_i)$ and that $\operatorname{Rad}(A)$ is a nilpotent ideal. Pick a basis of M_i, and let $e_{ij}^0 = E_{jj}^i$, the rank 1 projectors projecting to the basis vectors of this basis ($j = 1, \ldots, \dim M_i$). Then e_{ij}^0 are orthogonal idempotents in $A/\operatorname{Rad}(A)$. So by Corollary 9.1.3 we can lift them to orthogonal idempotents e_{ij} in A. Now define $P_{ij} = Ae_{ij}$. Then $A = \bigoplus_i \bigoplus_{j=1}^{\dim M_i} P_{ij}$, so P_{ij} are projective. Also, we have $\operatorname{Hom}(P_{ij}, M_k) = e_{ij} M_k$, so $\dim \operatorname{Hom}(P_{ij}, M_k) = \delta_{ik}$. Finally, P_{ij} is independent of j up to an isomorphism, as e_{ij} for fixed i are conjugate under $A^\times$ by Proposition 9.1.1; thus we will denote P_{ij} by P_i.

We claim that P_i is indecomposable. Indeed, if $P_i = Q_1 \oplus Q_2$, then $\operatorname{Hom}(Q_l, M_j) = 0$ for all j either for $l = 1$ or for $l = 2$, so either $Q_1 = 0$ or $Q_2 = 0$.

Also, there can be no other indecomposable finitely generated projective modules, since any indecomposable projective module has to occur in the decomposition of A. The theorem is proved. □

Definition 9.2.2. The projective module P_i is called the **projective cover** of M_i.

Proposition 9.2.3. *Let N be any finite dimensional A-module. Then one has* $\dim \operatorname{Hom}_A(P_i, N) = [N : M_i]$, *the multiplicity of occurrence on M_i in the Jordan-Hölder series of N.*

Proof. If $N = M_j$, the statement is clear. Also, if

$$0 \to N_1 \to N_2 \to N_3 \to 0$$

is an exact sequence of A-modules, then the corresponding sequence

$$0 \to \operatorname{Hom}_A(P_i, N_1) \to \operatorname{Hom}_A(P_i, N_2) \to \operatorname{Hom}_A(P_i, N_3) \to 0$$

is exact, as P_i is projective. This implies the statement. □